

MUSKEGON CGS, MI, USAWMO: **997341**Lat: **43.227N**Lon: **86.339W**Elev: **607**StdP: **14.38**Time Zone: **-5.00 (NAE)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.1	9.9	-1.7	5.1	7.0	2.0	6.2	11.3	38.6	22.0	35.1	24.9	15.6	320	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	79.9	71.2	77.9	70.4	76.2	69.4	74.0	77.7	72.4	76.0	71.0	74.7	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	123.2	76.9	71.0	116.9	75.2	69.6	111.2	73.9	37.8	77.9	36.4	76.2	35.2	74.6	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
34.7	30.9	27.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			0.6	83.7	5.7	2.3	-3.5	85.3	-6.8	86.6	-10.0	87.9	-14.1	89.5		
			0.5	75.4	6.3	2.2	-4.1	77.0	-7.8	78.2	-11.3	79.5	-15.9	81.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	25.1	24.4	32.0	44.6	54.4	63.9	69.0	68.7	62.6	51.3	41.2	32.5
	DBStd	17.26	8.59	7.71	7.88	6.95	7.27	5.70	5.47	4.58	5.81	6.86	7.51	7.18
	HDD50	3162	772	717	560	191	36	0	0	0	1	65	277	543
	HDD65	6746	1237	1137	1024	613	339	88	26	19	111	429	715	1008
	CDD50	2280	0	0	1	27	172	417	588	580	379	105	11	0
	CDD65	393	0	0	0	0	10	55	150	134	40	4	0	0
	CDH74	909	0	0	0	0	33	118	415	290	52	1	0	0
	CDH80	52	0	0	0	0	2	12	23	14	1	0	0	0
Wind	WSAvg	13.0	16.0	14.1	12.6	13.4	11.9	10.5	10.1	10.0	11.3	14.3	16.2	15.4
Precipitation	PrecAvg	35.0	2.2	1.8	2.3	3.4	3.5	3.5	3.0	3.1	3.1	3.5	3.1	2.3
	PrecMax	43.4	5.1	3.6	4.9	7.5	9.4	6.5	5.5	5.7	6.4	8.2	7.3	6.1
	PrecMin	27.8	0.6	0.7	0.6	0.7	0.5	1.3	1.2	1.0	0.4	0.3	0.6	1.0
	PrecStd	4.8	1.0	0.9	1.3	1.4	1.9	1.3	1.2	1.3	1.3	1.9	1.7	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	42.7	57.9	67.6	78.6	81.8	82.4	81.9	78.3	73.0	60.4	53.2
		MCWB	41.8	40.9	46.3	54.0	65.7	67.3	73.9	71.4	68.3	66.2	53.4	51.8
	2%	DB	41.8	38.9	50.3	61.8	73.4	77.3	80.0	79.2	75.6	69.1	57.0	47.7
		MCWB	39.8	37.3	43.9	51.4	63.0	66.3	72.6	71.5	68.8	64.2	52.3	45.5
	5%	DB	39.6	36.5	45.6	58.1	68.8	74.4	78.1	77.2	73.5	65.7	54.3	44.7
		MCWB	37.6	34.6	40.6	50.2	59.5	66.3	71.7	70.5	68.0	61.1	50.2	42.6
	10%	DB	37.3	34.4	42.0	54.8	65.0	72.0	76.4	75.5	71.6	62.0	51.8	42.2
		MCWB	35.3	32.5	37.9	47.9	57.5	65.2	70.6	69.2	66.2	57.5	47.8	40.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	42.1	49.0	56.5	68.4	71.4	76.4	75.6	72.3	68.2	55.8	52.3
		MCDB	43.1	42.8	51.8	59.9	76.5	76.8	80.8	79.2	75.6	71.3	58.7	53.2
	2%	WB	40.5	37.7	44.9	53.5	64.3	69.2	74.7	73.5	70.9	64.8	52.7	45.7
		MCDB	41.4	38.7	49.0	58.5	71.5	74.0	78.6	77.3	74.1	67.2	56.3	47.1
	5%	WB	37.8	35.0	41.2	50.3	61.9	67.7	73.0	72.1	69.4	60.8	50.4	42.8
		MCDB	39.3	36.1	45.5	55.6	67.7	72.4	76.6	75.6	72.9	64.2	53.4	44.3
	10%	WB	35.2	32.6	38.4	47.8	59.3	66.2	71.4	70.7	67.3	57.9	48.2	40.4
		MCDB	37.2	33.9	41.6	53.1	63.9	70.5	75.2	74.2	70.1	61.4	51.3	42.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.9	11.0	12.1	12.3	11.0	9.2	10.0	11.1	10.2	9.4	7.7
		MCDBR	10.0	11.1	14.8	17.7	17.3	13.9	10.4	11.1	12.0	12.2	11.3	10.5
		MCWBR	9.8	11.3	10.4	10.4	10.4	8.3	6.4	7.1	7.6	10.2	8.7	10.3
	5% WB	MCDBR	9.4	11.5	13.2	14.9	15.7	10.9	8.8	7.9	8.7	12.1	10.6	10.2
		MCWBR	10.0	11.7	10.2	10.9	11.6	7.3	6.0	6.6	7.0	10.9	9.2	10.2
Clear-Sky Solar Irradiance	taub	0.298	0.325	0.341	0.372	0.403	0.417	0.418	0.414	0.390	0.353	0.319	0.297	
	taud	2.421	2.330	2.375	2.320	2.301	2.312	2.313	2.329	2.360	2.450	2.486	2.463	
	Ebn at Noon	266	275	285	283	276	272	270	267	265	262	256	256	
	Edh at Noon	24	31	34	39	41	40	40	38	34	28	23	21	
All-Sky Solar Radiation	RadAvg	376	639	1131	1512	1779	1988	2023	1722	1367	840	483	303	
	RadStd	44	79	110	139	160	128	96	104	115	103	72	45	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page